

2.4.2 Intermediate Value Theorem Worksheet

1. Show precisely how the *Intermediate Value Theorem* is used to prove that the equation $x^3 - 2x - 3 = 0$ has at least one real solution.
2. Show that $f(x) = \sqrt[3]{x} - 1 + x$ has at least one real root.
3. Show that the equation $\cos(x) = x^3$ has at least one real solution.

4. Show that the equation $\ln(x) = 3 - 2x$ has at least one real solution.

5. Is there a number that is exactly 1 more than its cube? Use the IVT to answer this question.